

Platform

Project Overview

We want to develop a mobile application that uses **Augmented Reality (AR)** to help students learn human anatomy in an interactive and engaging way. Instead of studying anatomy through static textbook images, users can visualize and interact with **3D organs and body systems** using their smartphone camera.

The application should be affordable, user-friendly, and targeted toward medical, nursing, and biology students.

Main Objective

Create an AR-powered educational platform where users can:

- View 3D anatomical models in real-world space.
 - Learn anatomy through interactive exploration.
 - Access saved learning progress and previously viewed content.
 - Test their knowledge using AI-generated quizzes.
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Core Features

1. AR Anatomy Viewer

- Camera-based AR experience.
- Marker-based or markerless AR support.
- Display 3D organs such as:
 - Heart
 - Brain
 - Lungs
 - Kidney
 - Skeleton
- Rotate, zoom, and move models.

2. Interactive Labels

- Tap on an organ part to view:
 - Name
 - Function
 - Related body system
- Show educational descriptions and diagrams.

3. Layered Anatomy View

Users can switch between:

- Skeletal System
 - Muscular System
 - Nervous System
 - Digestive System
 - Circulatory System
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AI Features

AI Anatomy Assistant

A chatbot integrated within the app that can answer questions such as:

- "What is the function of the liver?"
- "Explain blood circulation."
- "What are the parts of the human heart?"

AI Quiz Generator

Generate quizzes automatically based on:

- Selected organ
- Selected body system
- User learning history

Quiz Types:

- Multiple Choice Questions
- True/False
- Fill in the blanks
- Image-based identification

Smart Difficulty Adjustment

- Beginner
- Intermediate
- Advanced

Difficulty changes according to user performance.

Learning Progress Tracking

User Dashboard

Store:

- Completed topics
- Quiz scores
- Time spent learning
- Learning streaks

Previous Data Storage

Users can access:

- Previously viewed organs
- Quiz history
- Saved notes
- Bookmarked anatomy models

Database Options:

- Firebase Firestore
 - Supabase
 - SQLite (offline support)
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Additional Features

Notes Section

Users can:

- Create notes for each organ
- Save important points
- Export notes as PDF

Bookmark System

Save favorite organs and topics for quick access.

Voice Explanation

- Text-to-speech for anatomy descriptions.
- Useful for accessibility and self-learning.

Search Functionality

Search:

- Organs
 - Systems
 - Medical terms
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Admin Panel (Optional)

- Upload new 3D models.
 - Add educational content.
 - Create custom quizzes.
 - Monitor user activity.
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Suggested Tech Stack

Frontend

- Flutter (Android & iOS)
or
- React Native

AR Development

- Unity + Vuforia
or
- ARCore (Android)
- ARKit (iOS)

Backend

- Firebase
- Supabase

AI Integration

- OpenAI API / Gemini API
- Quiz generation using AI models

Database

- Firestore
 - SQLite for offline storage
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Expected Outcome

A modern educational mobile application that combines **AR visualization, AI-powered learning assistance, progress tracking, previous data storage, and intelligent quizzes** to make anatomy learning more engaging, interactive, and effective for students.